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| **Fiscal Year:** | FY 2012 | **Task Last Updated:** | FY 03/08/2012 |
| **PI Name:** | Tseng, Bertrand Paul M.D., Ph.D. | | |
| **Project Title:** | Oxidative Stress and Charged Particle Irradiation Alter Multipotent Stem Cells to Elicit Acute and Functional Changes in Target Organ Systems | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | NSBRI | | |
| **Program/Discipline--Element/Subdiscipline:** | NSBRI--Radiation Effects Team | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 92697-2695 | **Congressional District:** | 48 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2008 NSBRI-RFA-08-03 Postdoctoral Fellowships |
| **Start Date:** | 11/01/2008 | **End Date:** | 10/31/2011 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NSBRI |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Limoli, Charles ( MENTOR/University of California, Irvine ) | | |
| **Grant/Contract No.:** | NCC 9-58-PF01901 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | POSTDOCTORAL FELLOWSHIP<br>Multipotent stem cells exist throughout the body and maintain the health of a given tissue by replenishing cells lost or damaged throughout life. These unique cells retain the capability to proliferate, migrate, and differentiate to return functionality to a tissue compromised by age, disease or a variety of geno/cytotoxic stressors such as ionizing radiation. Previously we showed that relatively low doses of radiation found in space (protons, heavy ions) can compromise the function of specific tissues and organs through a variety of biochemical mechanisms that involve oxidative stress. Oxidative stress that persists after irradiation can reduce bone density, impair muscle regrowth and inhibit neurogenesis. These acute, functional decrements could severely compromise the capability of astronauts to complete mission critical tasks, where relatively mild exposures to the space radiation environment may accelerate muscle fatigue and confusion | | |

**Task Description:**

caused by muscle and neural stem cell loss. These debilitating conditions can, in part, be caused by radiation-induced oxidative stress. Our data suggest that these multipotent cells may represent critical targets for countermeasures aimed at ameliorating radiation and space flight related sequelae. Thus, we propose in vitro and in vivo experiments to determine if and how relatively low doses of protons can initiate radioadaptive changes in stem cells found in the brain and skeletal muscle. Our proposal will evaluate how low dose proton exposure will impact stem cell proliferation, differentiation and survival and whether such endpoints can be altered by prior exposure. We hypothesize that stem cells incurring particle traversals during spaceflight can undergo adaptive changes that attenuate their response to subsequent larger exposures (e.g. from a solar flare). We will also investigate whether the mechanistic basis of adaptation involves changes in the redox environment such as oxidative stress.

Three aims are proposed:

Specific Aim 1: To determine if/how low dose proton irradiation elicits radioadaptive changes in neural precursor cells from the CNS that depend on oxidative stress.

Specific Aim 2: To determine the acute dose response of satellite cells to proton irradiation with respect to survival, phenotypic fate, and oxidative stress.

Specific Aim 3: Elucidate how low dose irradiation impacts neurogenesis in mice expressing human catalase targeted to the mitochondria (MCAT).

In aims 1 and 2, a range of biochemical, oxidative, and immunohistochemical approaches will be used that involve the quantification of reactive oxygen and nitrogen species and the detection of oxidative damage in neural precursor and satellite cells. In aim 3, we will utilize mice genetically modified to minimize mitochondrial hydrogen peroxide to determine how reducing intracellular oxidants impacts neurogenesis. These studies will help establish whether adaptation occurs after low dose proton irradiation, and whether stem cells and oxidative stress represent logical targets for countermeasure strategies.

The key findings are that low dose proton irradiation of rodent and human derived NPCs and rodent satellite cells results in ROS and RNS levels that fluctuate over time as well as a gradual increase in superoxide levels. Further, both dose and dose rate may be key components of radioadaptation in the multipotent cells when examining survival/proliferation using our priming-challenge paradigm.

The main impact of these finding is that we will focus on better understanding the radioadaptive effect of low dose and low dose rate proton irradiation on the NPCs and satellite cells. The plan for the year was to determine the dose limits and dose rate dependence of the priming dose with respect to eliciting radioadaptation and the temporal interval during which the cells are radioadapted. Lastly, work continues in determining whether or not the MCAT mice are radioprotected with respect to neurogenesis.

**Rationale for HRP Directed Research:**

**Research Impact/Earth Benefits:**

Very little is known about the low dose and low dose rate radiobiology of neural precursor cells. These studies will yield important information regarding how this radioresponse differs from the response to higher doses of ionizing radiation. Further, increased reactive oxygen species (ROS) can be damaging by challenging cellular repair and renewal processes but it can also activate redox sensitive signaling pathways which may underlie an adaptive response in neural precursors. Our recent work has begun to elucidate the effectiveness of low dose rate irradiation on the radioadaptation of neural precursor cells and it may prove to be quite beneficial as these dose rates better simulate the space radiation environment. Knowledge gained from understanding the mechanisms of change in oxidative stress, mitochondrial function and redox sensitive signaling in these experiments will provide the rationale for potentially effective countermeasure design for both acute and chronic space radiation exposure.
Ionizing radiation significantly reduces dentate neurogenesis, and such changes are dose dependent and persistent. These effects are linked to alterations in the neurogenic microenvironment, including inflammatory changes and oxidative stress although the precise mechanisms responsible are not yet known. While ROS have often been considered to be hostile or destructive entities they also have been shown to have beneficial effects, at least in part due to their role as signaling moieties. It has recently been shown that a persistent level of oxidative stress in extracellular superoxide dismutase knockout mice (EC-SOD KO) was associated with a lower baseline level of neurogenesis relative to wildtype (WT) mice. However, when those same mice were subjected to a modest dose of x-rays (5 Gy), there was no effect on neurogenesis in KO mice but a significant reduction in WT mice. Thus, we saw both negative (baseline neurogenesis) and positive (adaptive) effects in the EC-SOD KO mice, presumably as a result of mechanisms associated with redox balance. Also we have also been able to demonstrate the paradoxical nature of oxidative species in vitro using cultured neural precursor cells, where excess hydrogen peroxide reduces survival while excess superoxide increases survival. These paradoxical effects highlight the potential importance of adaptive responses in the context of the delicate balance in redox homeostasis, and how that may ultimately affect cell or tissue function. We believe that low doses of irradiation will elevate the level of oxidative stress in the neurogenic microenvironment and that this may have a beneficial effect when buffered by enhanced catalase activity with respect to dentate neurogenesis. These experiments can be conducted using a transgenic mouse with targeted expression of human catalase in the mitochondria (MCAT). Given the association between neurogenesis and cognitive function after irradiation, this modulation of oxidative stress via catalase could ultimately be developed into a countermeasure to maintain behavioral performance while engaged in space exploration.

With the capability to culture satellite cells in a multipotent state, we can now investigate the mechanistic details of ROS and RNS production in live cells. Changes in ROS and RNS following proton irradiation may prove be similar to what we have observed in either neural precursors exposed to the same irradiation at the same doses suggesting a common radioresponse pathway or quite different suggesting a more cell type specific response. Similarly mitochondrial redox function via superoxide levels can be readily assayed in these. Understanding these effects of irradiation on myogenic oxidative stress, proliferation and differentiation will help to provide the basis in order to design countermeasures to maintain muscle function during space exploration.

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| **Task Progress:** | To examine if lower priming doses could elicit a similar improvement in survival to our 25 cGy priming doses, we assay for survival using 10 cGy proton priming doses. These 10 cGy proton priming for hNSCs 24h prior to 1 Gy gamma challenge showed a non-statistically significant improvement in survival after 5 days post challenge of between 1-2% for both HDR and LDR priming dose rates. Experiments with hNSCs were also conducted to examine if changing the interval between priming and challenge would show differences in post challenge survival. Unfortunately, due to the difficult technical nature of using these cells, multiple attempts to generate data from a 5-6h priming challenge interval resulted in premature cell demise prior to the usual 5d post-challenge assay time point. Whether this was a result of the transitions to and from the irradiations in a such a short time period resulting in changes in temperature, pH, and CO2 levels that may have been detrimental to the survival of these cells is still unknown although similar survival problems has been observed with these cells when used for iron irradiations at Brookhaven National Lab. Thus rodent neurospheres were examined using a 6h time interval as they tend to be a bit more robust in terms of experimental handling. The results were similar to the hNSCs for the 10 cGy HDR proton priming followed by 2 Gy gamma challenge in that a non-statistically significant improvement of 3-4% was observed for 5 day post-challenge survival.<br>Rodent satellite cells were also examined in a similar time interval paradigm of HDR proton priming, then gamma challenge with the addition of a 12h time interval. These cells again appeared to be relatively radio-resistant to a 2 Gy challenge dose and for the 6h time interval, a 1-2% non-statistically significant improvement in survival was observed whereas no improvement was seen for the 12h time interval following a 25 cGy HDR proton priming dose. A manuscript containing the results of the oxidative stress and survival studies conducted using the rodent and human neural stem/precursor cells was submitted. Interesting results were also observed for the post challenge oxidative stress levels in the priming/challenge paradigm and these results are discussed in more detail in the main findings section.<br><br>Lastly, work continues on the MCAT aim as brain sections are in the process of being immunostained for BrdU and NeuN for the 6 groups of 8 animals per group with a target of 10-15 sections per animal which are then analyzed by confocal microscopy for the co-localization of the BrdU and NeuN. |
| **Bibliography Type:** | Description: (Last Updated: 08/12/2013) |
| **Articles in Peer-reviewed Journals** | Cho-Lim JJ, Caiozzo VJ, Tseng BP, Giedzinski E, Baker MJ, Limoli CL. "Satellite cells say NO to radiation." Radiat Res. 2011 May;175(5):561-8. Epub 2011 Feb 14. PMID: 21319985 , May-2011 |
| **Articles in Peer-reviewed Journals** | Tseng BP, Lan ML, Tran KK, Acharya MM, Giedzinski E, Limoli CL. "Characterizing low dose and dose rate effects in rodent and human neural stem cells exposed to proton and gamma irradiation." Redox Biol. 2013;1(1):153-62. http://dx.doi.org/10.1016/j.redox.2013.01.008 , Jul-2013 |